1 What is Flutter?

Ans ->Flutter is an open-source UI software development toolkit created by Google. It is used to build cross-platform applications for Android, iOS, Web, Windows, macOS, and Linux using a single codebase.

Key Features of Flutter

Single codebase for multiple platforms

Fast development with Hot Reload

Beautiful and customizable UI

High performance because it compiles into native code

Uses the Dart programming language

2 What is Dart and Why does Flutter use it?

Ans -> Dart is an object-oriented programming language developed by Google. It is mainly used to build mobile, web, desktop, and server applications.

Dart can compile directly into native machine code using AOT (Ahead Of Time) compilation.

3 What is pubspec.yaml file and what does it do?

Ans-> In Flutter and Dart, the pubspec.yaml file is the main configuration file of the project.

It contains important information about the app such as:

Project name

Version

Dependencies (packages)

Assets

Fonts

Environment settings

4 What is the difference between main() and runApp() functions in Flutter?

Ans -> main() is the entry point of every Dart application.

Execution of the app starts from here.

ex:void main() {

print(“App Started”);

}

runApp() Function

runApp() is a Flutter function used to display the root widget on the screen.

It takes a widget and attaches it to the screen.

ex: runApp(MyApp());

5 Differentiate between named parameters and positional parameters in Flutter.

Ans-> 1. Positional Parameters

Parameters are passed based on their position/order.

ex: void showDetails(String name, int age) {

print(“Name: $name”);

print(“Age: $age”);

}

void main() {

showDetails(“Nitish”, 20);

}

2. Named Parameters

Parameters are passed using parameter names

ex: void showDetails({String? name, int? age}) {

print(“Name: $name”);

print(“Age: $age”);

}

void main() {

showDetails(name: “Nitish”, age: 20);

}

6 What are widgets in Flutter?

Ans-> a Widget is the basic building block of the user interface (UI).

Examples:Text,Button,Image,Row,Column,Screen layout

7 What do you mean by open-source software? Is Flutter open-source?

Ans-> Open-source software is software whose source code is publicly available so that anyone can use, modify, and distribute it. Yes, Flutter is an open-source UI framework developed by Google.

8 Differentiate between StatelessWidget and StatefulWidget in Flutter.

Ans->

| StatelessWidget | StatefulWidget |

A widget whose data cannot change after the UI is built. | A widget whose data can change during runtime. |

Used for static UI. | Used for dynamic UI. |

Does not use `setState()`. | Uses `setState()` to update UI. |

Faster and lightweight. | Slightly heavier because state management is involved. |

Example: Text, Icon | Example: Checkbox, Counter, Form |

9 What is BuildContext in Flutter? And why is it needed?

Ans ->BuildContext is a handle to a widget's location in the element tree, identifying where it sits relative to other widgets.

It is required to look up data from ancestors (like themes or navigation) and to manage the widget's relationship with the underlying render tree.

10 What is the difference between WidgetsApp and MaterialApp in Flutter?

Ans-> WidgetsApp is a basic Flutter app configuration widget that provides essential features like navigation, routing, and localization without Material Design.

MaterialApp is built on top of WidgetsApp and adds Material Design features such as themes, AppBar, Scaffold, and Material UI components in Flutter.

11 Differentiate between final, const and static keyword.

Ans->final is used to assign a value only once at runtime, const is used for compile-time constant values, and both cannot be changed after initialization.

static is used to create class-level variables or methods that belong to the class itself rather than to an object in Dart.

12 What are Slivers?

Ans-> Slivers are scrollable areas used to create custom scrolling effects and advanced scrollable layouts.

They are mainly used inside CustomScrollView for features like collapsing App-Bar, lazy loading lists, and flexible scrolling UI.

13 What are mixins? How to use them?(dart qution).

Ans-> mixins are a way to reuse code in multiple classes without using inheritance.

They are used with the with keyword to add properties and methods of one class into another class.

14 What do you mean by Null-aware operators?

Ans-> null-aware operators are used to handle null values safely and avoid null reference errors.

Common null-aware operators are ?., ??, ??=, and !, which help check or assign values when an object may be null.

15 What is an AspectRatio widget used for?

Ans-> the AspectRatio widget is used to maintain a fixed width-to-height ratio for its child widget.

It helps create responsive UI layouts by automatically adjusting the child's size according to the given aspect ratio.

16 What are DevTools in Flutter?

Ans-> DevTools are a set of performance and debugging tools used to inspect, analyze, and optimize Flutter applications.

They help developers monitor UI rendering, memory usage, network activity, app performance, and widget structure during development

17 How would you make HTTP requests in the Flutter framework?

Ans->HTTP requests are commonly made using the http package to communicate with APIs or servers.

Methods like GET, POST, PUT, and DELETE are used to send and receive data asynchronously in Flutter applications.

18 What is the difference between Provider vs. InheritedWidget?

Ans-> InheritedWidget is a built-in Flutter widget used to share data efficiently down the widget tree.

Provider is a higher-level state management solution built on top of InheritedWidget that makes state handling simpler, cleaner, and easier to manage in Flutter.

16 What are the differences between JIT and AOT?

Ans->JIT (Just-In-Time) compilation converts code during runtime and is mainly used for faster development features like Hot Reload in Flutter.

AOT (Ahead-Of-Time) compilation converts code into native machine code before app execution, providing better performance and faster startup in release mode.

17 How do Object, dynamic, and var differ in Dart?

Ans-> Object can store any type of value but allows access only to methods available in the Object class, while dynamic can store any type and allows calling any method without compile-time checking.

var automatically infers the variable type from the assigned value at compile time in Dart.

18 List the responsibilities of FlutterActivity.

Ans->FlutterActivity is responsible for launching and displaying a Flutter screen inside an Android application.

It manages the connection between Android and the Flutter engine, including app lifecycle, rendering, plugin registration, and communication with platform-specific code in Flutter.

19 What is a MediaQuery in Flutter?

Ans->MediaQuery is used to get information about the device screen, such as width, height, orientation, and pixel density.

20 What are the advantages of a Flutter Inspector?

Ans->The Flutter Inspector in Flutter helps developers visualize and debug the widget tree of a Flutter application.

Its advantages include easier UI debugging, layout inspection, performance analysis, and quick identification of widget-related issues during development.

21 What is difference bt APK or AAB file

Ans-> APK (Android Package Kit) is a complete Android app file used for direct installation on devices(flutter build apk)

AAB (Android App Bundle) is a publishing format used by Google Play Store to generate optimized APKs for different devices.(flutter build appbundle)

//========== dart qutions=======

1 what is dart

Ans-> Dart is an open-source, object-oriented programming language developed by Google for building mobile, web, desktop, and server applications.

It is mainly used with Flutter to create fast and cross-platform applications.

2 What is a pub in Dart?

Ans-> pub is the package manager used to manage dependencies, libraries, and project packages.

It helps developers install, update, and organize external packages in Dart and Flutter projects.

3 Explain polymorphism in Dart.

Ans-> polymorphism is an object-oriented programming concept where the same method or object can behave differently based on the class implementation.

It is mainly achieved through method overriding and inheritance, allowing code reusability and flexibility.

4 When can you use generics?

Ans->Generics are used when writing reusable and type-safe code that can work with different data types.

They help reduce code duplication and improve compile-time type checking in classes, methods, and collections.

5 What is the use of @Deprecated annotation in Dart?

Ans-> @Deprecated is used to mark old code that should not be used anymore.

It gives a warning to developers to use new or updated code instead.

6 What is the difference between final and const?

Ans-> final variables are assigned only once at runtime, while const variables are fixed at compile time.

Both cannot be changed after initialization, but const values must be completely constant before the app

7 How do you catch an error in the Dart program?

Ans->, errors are caught using try, catch, and finally blocks.

The try block contains risky code, and the catch block handles the error if an exception occurs.

8 What are "async", "await"and "Future"?

Ans->Future represents a value that will be available later, async is used to make a function asynchronous, and await pauses execution until the Future is completed.

They are mainly used for tasks like API calls, database operations, and file handling without blocking the app

9 What are the differences between Object and dynamic?

Ans-> Object can store any type of value but only allows access to methods defined in the Object class.

dynamic can also store any type of value but allows calling any method without compile-time type checking.

10 What is assert in dart?

Ans->assert is used to check whether a condition is true during development.

If the condition is false, it throws an error and helps developers detect bugs early.

11 What is Cascade notation in dart?

Ans->cascade notation (..) is used to perform multiple operations on the same object without repeating the object name.

It makes the code shorter, cleaner, and more readable.

12 What is the spread operator in dart?

Ans-> the spread operator (…) is used to insert multiple elements of one collection into another collection.

It helps combine lists, sets, and maps easily in a clean and readable way.

13 What is isolate?

Ans-> an isolate is an independent worker that runs code separately from the main thread.

It is used to perform heavy tasks in the background without blocking the UI or app performance.

14 How does Dart/Flutter get compiled to native android or ios?

Ans-> Dart code in Flutter is compiled into native machine code using AOT (Ahead-Of-Time) compilation for Android and iOS.

This allows Flutter apps to run fast and efficiently without needing a separate JavaScript bridge.

15 Does dart uses VM, if so, how?

Ans->Yes, Dart uses the Dart VM (Virtual Machine) during development.

The Dart VM supports JIT (Just-In-Time) compilation, which enables fast development features like Hot Reload in Flutter.